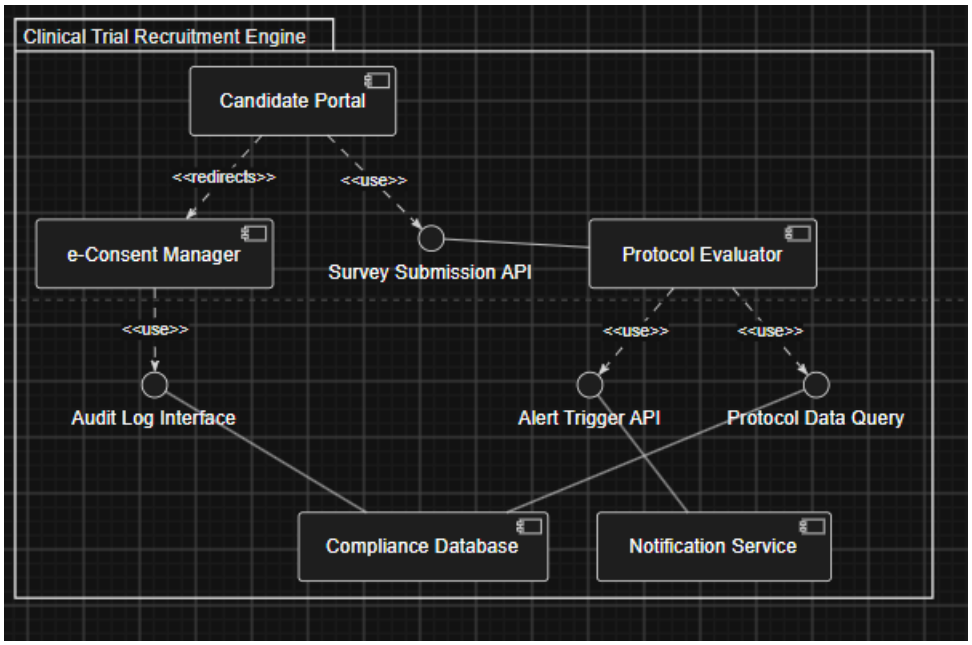
Lab 3

PES2UG24CS386

R G Nithik

Component Diagram for The Clinical Trial Recruitment Engine:



Architecture Selection: "We chose Microservices Architecture for the Clinical Trial Recruitment Engine System."

Architectural Choice:

Microservices Architecture

Two Reasons:

1. Scalability for Performance Constraints: The system must evaluate candidate survey responses against eligibility criteria in under 3 seconds, as specified in NFR-002. The independent scaling capability of microservices allows the evaluation engine to scale automatically during high traffic without dedicating unnecessary resources to the rest of the application.

2. Fault Isolation: Microservices offer excellent fault isolation. If the notification system fails while dispatching automated alerts (FR-004), candidates will still be able to submit their

eligibility surveys and sign their e-Consent forms without experiencing system-wide downtime.

Security Advantage:

This architecture supports technology diversity. This allows the development team to use highly specialized, secure cryptographic frameworks exclusively for the e-Consent module to ensure mandatory compliance with FDA 21 CFR Part 11 (NFR-001). This isolates the sensitive audit trail data from general system components, minimizing the overall attack surface and strictly controlling access to patient compliance records.

Performance Benefit:

Because microservices are independently deployable, the system can allocate dedicated processing power specifically to the protocol evaluation service. This ensures the 3-second evaluation limit (NFR-002) is strictly maintained without being bottlenecked by slower or resource-intensive operations, such as logging subject visit compliance (FR-003) or managing custom trial protocols (FR-005).